

# Quantum Communication in Weakly Turbulent Channels: Error Analysis and Capacity Bounds

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**Abstract**—In this letter, we analyze the average probability of error of quantum modulation schemes in weakly turbulent quantum communication channels utilizing coherent states. Atmospheric turbulence-induced fading is a significant challenge that degrades the performance of free-space quantum communication systems. We derive and present closed-form expressions for the average probability of error for various detection methods: classical (shot-noise limited) single-photon detection, coherent detection, and the optimal quantum hypothesis-testing (Helstrom) detection for binary quantum modulation schemes. Our analysis focuses on the transmission of classical data modulated by quantum coherent states over a log-normal turbulent channel. Additionally, we discuss the lower bounds of the average capacities for Helstrom detection, providing comprehensive insights into the achievable information rates in realistic atmospheric conditions. The derived expressions show excellent agreement with numerical Monte Carlo simulations across a broad range of average transmitted photon numbers, offering valuable guidelines for the design and optimization of quantum communication systems.

**Index Terms**—Quantum communication, error probability, turbulence, quantum modulation, lognormal fading.

## I. INTRODUCTION

QUANTUM communication systems, based on quantum mechanics principles, promise to transform secure and efficient data transmission [1], [2], [3]. Quantum-enhanced classical schemes can reduce error probabilities compared to classical methods by employing optimal quantum receivers [4]. Coherent states for classical data transmission in free-space quantum systems have gained significant interest [5], [6]. However, these systems face challenges from atmospheric

turbulence, which causes scintillation, beam wandering, and wavefront distortion, leading to increased error rates. Addressing these effects is crucial for developing robust quantum communication systems.

Several studies have explored the impact of turbulence on quantum communications [5], [7], [8]. For instance, [9] analyzed Shannon capacities for classical and quantum detection with binary discrete modulations, on-off keying (OOK) and binary phase-shift keying (BPSK) by assuming a pure-loss channel. However, a comprehensive analysis of average error probability in weakly turbulent channels with discrete modulation of coherent states is lacking. Such analysis is vital for optimizing quantum communication systems under realistic conditions. The authors of [10] investigated turbulence effects with BPSK modulation and proposed using receiver diversity and generalized Kennedy receivers to mitigate turbulence and thermal noise. Their focus was on reducing error probabilities and comparing receiver variants.

This letter provides a comprehensive analysis by deriving closed-form expressions for average error probability in weakly turbulent quantum channels. Unlike [10], which centers on Kennedy receivers, we examine three detection schemes: classical (shot-noise limited) single-photon detection, coherent detection, and the optimal quantum hypothesis-testing (Helstrom) detection. Using a log-normal model for turbulence-induced fading, our study thoroughly assesses these methods' performance under realistic conditions. The derived analytical expressions are validated against numerical Monte Carlo simulations, demonstrating their accuracy over a wide range of average transmitted photon numbers. We also explore the lower bounds of average capacities for Helstrom detection, offering insights into achievable information rates in practical scenarios.

## II. SYSTEM MODEL

We consider a free-space optical quantum communication channel, where coherent states emitted from a laser source serve as the carrier for information transmission. Coherent states, denoted by  $|\alpha\rangle$  are used as information carriers, where  $\alpha$  is a complex number. These states can be described using the complete orthonormal set in the Fock basis  $|n\rangle$  ( $n = 0, 1, 2, \dots$ ) as  $|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ . Let the average number of transmitted photons be  $N_s = |\alpha|^2$ . In an ideal channel, the average number of received photons would be  $E = N_s$ .

For each channel use, the transmitter prepares a coherent state modulated by the random variable  $X \in \mathcal{X} = \{0, 1\}$ , with  $X = 0$  occurring with probability  $p_0$  and  $X = 1$  with probability  $p_1$ . In the case of OOK, the transmitted signal is either  $|\alpha_0\rangle = |0\rangle$  for bit 0 or  $|\alpha_1\rangle = |\alpha\rangle$ , for bit 1. For BPSK, the two

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states are  $|\alpha_0\rangle = |-\alpha\rangle$  and  $|\alpha_1\rangle = |\alpha\rangle$ , representing bits 0 and 1 respectively. In a turbulent channel, the received signal can be modeled as  $\{|\sqrt{\eta_{eq}}\alpha_0\rangle, |\sqrt{\eta_{eq}}\alpha_1\rangle\}$ , where  $\eta_{eq} = \eta_t \eta_{sc}$  is the equivalent channel transmissivity. Here,  $\eta_t$  represents the dynamic effects of turbulence, while  $\eta_{sc}$  accounts for static scattering and absorption losses, which remain relatively constant over short intervals. This decomposition of  $\eta_t$  and  $\eta_{sc}$  provides a more accurate characterization of the distinct physical effects on signal propagation in free-space quantum channels [8], [11]. Thus, for both modulation schemes, the average number of received photons can be expressed as  $E = p_0 \eta_{eq} |\alpha_0|^2 + p_1 \eta_{eq} |\alpha_1|^2$ .

### III. PROBABILITY OF ERROR IN OOK

#### A. Direct Detection With Photon Counting

In OOK (with equiprobable symbols) with direct detection using photon counting (PC), the incoming photons are characterized by a Poisson distribution with a mean photon number  $N_s/2$  and the probability of error is given by  $P_{OOK}^{PC} = e^{-2E}/2$  [6, Ch. (7)]. When the coherent states  $|0\rangle$  and  $|\alpha\rangle$  are transmitted through a turbulent atmosphere, they become  $|0\rangle$  and  $|\sqrt{\eta_{eq}}\alpha\rangle$ , respectively, where the effective mean received photon number is  $E = \eta_{eq} N_s/2$ . The probability of error in the presence of turbulence is given by

$$P_{OOK}^{PC}(\eta_{eq}) = \frac{e^{-\eta_{eq} N_s}}{2}. \quad (1)$$

Considering weak-to-moderate turbulence, the transmissivity  $\eta_t$  follows a lognormal distribution with the probability density function (PDF) [12]

$$p(\eta_t) = \frac{1}{\eta_t \sqrt{2\pi\sigma_l^2}} \exp\left(-\frac{\left(\ln(\eta_t) + \frac{\sigma_l^2}{2}\right)^2}{2\sigma_l^2}\right), \quad (2)$$

where  $\sigma_l^2$  represents the Rytov parameter, which quantifies the variance of log-irradiance and thus indicates the strength of turbulence. Since  $\eta_{eq} = \eta_t \eta_{sc}$ , where  $\eta_{sc}$  is a constant, the distribution of  $\eta_{eq}$  can be derived by scaling  $\eta_t$  by  $\eta_{sc}$ . Therefore, if  $\eta_t$  is lognormally distributed, then  $\eta_{eq}$  will also be lognormally distributed, but with a different location parameter:

$$p(\eta_{eq}) = \frac{1}{\eta_{eq} \sqrt{2\pi\sigma_l^2}} \exp\left(-\frac{\left(\ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) + \frac{\sigma_l^2}{2}\right)^2}{2\sigma_l^2}\right). \quad (3)$$

The average error probability,  $\bar{P}_{OOK}^{PC}$ , can be found by integrating  $P_{OOK}^{PC}(\eta_{eq})$  over the PDF of  $\eta_{eq}$  as

$$\bar{P}_{OOK}^{PC} = \int_0^\infty P_{OOK}^{PC}(\eta_{eq}) p(\eta_{eq}) d\eta_{eq}. \quad (4)$$

Substituting the expression for  $P_{OOK}^{PC}(\eta_{eq})$

$$\begin{aligned} \bar{P}_{OOK}^{PC} &= \int_0^\infty \left(\frac{e^{-\eta_{eq} N_s}}{2}\right) \\ &\times \frac{1}{\eta_{eq} \sqrt{2\pi\sigma_l^2}} \exp\left(-\frac{\left(\ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) + \frac{\sigma_l^2}{2}\right)^2}{2\sigma_l^2}\right) d\eta_{eq}, \end{aligned} \quad (5)$$

$$\bar{P}_{OOK}^{PC} = \frac{1}{2} \mathcal{L}(N_s), \quad (6)$$

where  $\mathcal{L}(s)$  is the Laplace transform of the lognormal distribution with parameters  $\mu = \ln(\eta_{sc}) - \sigma_l^2/2$  and  $\sigma^2 = \sigma_l^2$ .

$$\mathcal{L}(s) = \int_0^\infty \frac{e^{-s\eta_{eq}}}{\eta_{eq} \sqrt{2\pi\sigma_l^2}} \exp\left(-\frac{\left(\ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) + \frac{\sigma_l^2}{2}\right)^2}{2\sigma_l^2}\right) d\eta_{eq}. \quad (7)$$

A closed-form expression of the Laplace transform of the lognormal distribution does not exist and can be evaluated by using numerical methods [13]. According to [14, eq. (5-7)], a simple closed-form expression for the expectation of any function of a normal random variable  $X \sim N(\mu_x, \sigma_x^2)$  with the PDF  $f_x(x)$  can be obtained as

$$\begin{aligned} E[g(x)] &= \int_{-\infty}^\infty g(x) f_x(x) dx \\ &\approx \frac{2}{3} g(\mu_x) + \frac{1}{6} g(\mu_x + \sqrt{3}\sigma_x) + \frac{1}{6} g(\mu_x - \sqrt{3}\sigma_x). \end{aligned} \quad (8)$$

Using this result, a simplified closed-form expression for the average error probability can be obtained. By substituting  $\ln(\frac{\eta_{eq}}{\eta_{sc}}) = z$  in (5), we get

$$\bar{P}_{OOK}^{PC} = \frac{1}{2} \int_{-\infty}^\infty \frac{e^{-\eta_{sc} N_s e^z}}{\sqrt{2\pi\sigma_l^2}} \exp\left(-\frac{\left(z + \frac{\sigma_l^2}{2}\right)^2}{2\sigma_l^2}\right) dz, \quad (9)$$

$$\bar{P}_{OOK}^{PC} = \frac{1}{2} \int_{-\infty}^\infty \varphi(z) f_z(z) dz, \quad (10)$$

where  $\varphi(z) = e^{-\eta_{sc} N_s e^z}$  and  $f_z(z)$  is the PDF of the normal random variable  $Z$  with mean  $\mu_z = -\sigma_l^2/2$  and variance  $\sigma_z^2 = \sigma_l^2$ . Therefore, an approximate expression of the average error probability can be written as

$$\bar{P}_{OOK}^{PC} \approx \frac{1}{2} \left[ \frac{2}{3} \varphi(\mu_z) + \frac{1}{6} \varphi(\mu_z + \sqrt{3}\sigma_z) + \frac{1}{6} \varphi(\mu_z - \sqrt{3}\sigma_z) \right], \quad (11)$$

which can be simplified as

$$\begin{aligned} \bar{P}_{OOK}^{PC} &\approx \frac{1}{2} \left[ \frac{2}{3} \exp(-\eta_{sc} N_s e^{-\sigma_l^2/2}) \right. \\ &\quad + \frac{1}{6} \exp(-\eta_{sc} N_s e^{-\sigma_l^2/2 + \sqrt{3}\sigma_l}) \\ &\quad \left. + \frac{1}{6} \exp(-\eta_{sc} N_s e^{-\sigma_l^2/2 - \sqrt{3}\sigma_l}) \right]. \end{aligned} \quad (12)$$

This result describes the impact of atmospheric turbulence strength on the average error probability. It can be observed that the average error probability increases as  $\sigma_l^2$  increases. A larger  $\sigma_l^2$  indicates stronger turbulence, leading to more significant fluctuations in the received signal, thereby increasing error probability. Minimizing the turbulence effects ( $\sigma_l^2$ ) and maximizing the transmitted photons ( $N_s$ ) are critical strategies for reducing the average error probability.

#### B. Helstrom Detection

The Helstrom detection limit, based on quantum hypothesis testing, provides the optimal resolvability of binary quantum states [15], [16]. The optimal detection error probability given by Helstrom bound for OOK is  $P_{OOK}^{\text{Hels}} = \frac{1}{2}(1 - \sqrt{1 - e^{-2E}})$  [9]. When considering turbulence, the effective mean received photon number becomes  $E = \eta_{eq} N_s/2$ , and

the Helstrom limit for the probability of error is given by

$$P_{\text{OOK}}^{\text{Hels}}(\eta_{eq}) = \frac{1}{2} \left( 1 - \sqrt{1 - e^{-\eta_{eq} N_s}} \right). \quad (13)$$

This formula provides the quantum limit for the probability of error in the presence of turbulence. Considering weak-to-moderate turbulence, the average error probability is given by

$$\bar{P}_{\text{OOK}}^{\text{Hels}} = \int_0^\infty P_{\text{OOK}}^{\text{Hels}}(\eta_{eq}) p(\eta_{eq}) d\eta_{eq}, \quad (14)$$

$$\bar{P}_{\text{OOK}}^{\text{Hels}} = \int_0^\infty \frac{1}{2} \left( 1 - \sqrt{1 - e^{-\eta_{eq} N_s}} \right) \times \frac{1}{\eta_{eq} \sqrt{2\pi\sigma_l^2}} \exp \left( -\frac{\left( \ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) + \frac{\sigma_l^2}{2} \right)^2}{2\sigma_l^2} \right) d\eta_{eq}. \quad (15)$$

Separating the terms and using  $\ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) = z$ , we have

$$\bar{P}_{\text{OOK}}^{\text{Hels}} = \frac{1}{2} - \frac{1}{2} \int_{-\infty}^\infty \sqrt{1 - e^{-\eta_{sc} N_s e^z}} \times \frac{1}{\sqrt{2\pi\sigma_l^2}} \exp \left( -\frac{\left( z + \frac{\sigma_l^2}{2} \right)^2}{2\sigma_l^2} \right) dz. \quad (16)$$

Using the result from [14, eq. (5-7)], an approximate expression for the average error probability can be written as

$$\bar{P}_{\text{OOK}}^{\text{Hels}} \approx \frac{1}{2} - \frac{1}{2} \left[ \frac{2}{3} \sqrt{1 - \exp(-\eta_{sc} N_s e^{-\sigma_l^2/2})} + \frac{1}{6} \sqrt{1 - \exp(-\eta_{sc} N_s e^{-\sigma_l^2/2 + \sqrt{3}\sigma_l})} + \frac{1}{6} \sqrt{1 - \exp(-\eta_{sc} N_s e^{-\sigma_l^2/2 - \sqrt{3}\sigma_l})} \right]. \quad (17)$$

The average error probability depends on  $\sigma_l^2$  and  $N_s$ . It increases with  $\sigma_l^2$  and decreases with  $N_s$ . Specifically, as  $\sigma_l^2$  grows, the term inside the exponential function increases, causing the square root term to approach zero, which raises the average error probability.

#### IV. PROBABILITY OF ERROR IN BPSK

##### A. Homodyne Detection

For BPSK-modulated signals, since both symbols have the same optical energy, direct detection using photon counting cannot discriminate between them. Therefore, coherent communication based on semiclassical theory employs homodyne measurements to detect BPSK signals. The error probability for BPSK signals with mean received photon number  $E = N_s$  is given by  $P_{\text{BPSK}}^{\text{Hom}} = \frac{1}{2} \text{erfc}(\sqrt{2E})$  [9]. When these states,  $|\alpha\rangle$  and  $|\alpha\rangle$ , propagate through a turbulent atmosphere, they become  $|\sqrt{\eta_{eq}}\alpha\rangle$  and  $|\sqrt{\eta_{eq}}\alpha\rangle$ , with the mean received photon number in each state being  $\eta_{eq} N_s$ . Thus, the effective mean received photon number becomes  $E = \eta_{eq} N_s$ , and the error probability under turbulence is given by

$$P_{\text{BPSK}}^{\text{Hom}}(\eta_{eq}) = \frac{1}{2} \text{erfc}(\sqrt{2\eta_{eq} N_s}). \quad (18)$$

The average error probability is given by

$$\bar{P}_{\text{BPSK}}^{\text{Hom}} = \int_0^\infty \frac{1}{2} \text{erfc}(\sqrt{2\eta_{eq} N_s}) \times \frac{1}{\eta_{eq} \sqrt{2\pi\sigma_l^2}} \exp \left( -\frac{\left( \ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) + \frac{\sigma_l^2}{2} \right)^2}{2\sigma_l^2} \right) d\eta_{eq}. \quad (19)$$

Using the result from [14, eq. (5-7)], the average error probability for BPSK Homodyne detection can be approximated as

$$\bar{P}_{\text{BPSK}}^{\text{Hom}} \approx \frac{1}{2} \left[ \frac{2}{3} \text{erfc} \left( \sqrt{2\eta_{sc} N_s e^{-\sigma_l^2/2}} \right) + \frac{1}{6} \text{erfc} \left( \sqrt{2\eta_{sc} N_s e^{-\sigma_l^2/2 + \sqrt{3}\sigma_l}} \right) + \frac{1}{6} \text{erfc} \left( \sqrt{2\eta_{sc} N_s e^{-\sigma_l^2/2 - \sqrt{3}\sigma_l}} \right) \right]. \quad (20)$$

It can be observed that as  $\sigma_l^2$  increases, the quantity inside the complementary error function decreases, leading to an increase in the average error probability. This is because the complementary error function,  $\text{erfc}(\cdot)$ , is a decreasing function; thus, a smaller argument results in a larger value of the function, increasing the overall error probability.

##### B. Helstrom Detection

The Helstrom limit for BPSK with mean received photon number  $E$  is given by  $P_{\text{BPSK}}^{\text{Hels}} = \frac{1}{2} (1 - \sqrt{1 - e^{-4E}})$ . With turbulence, the effective mean received photon number becomes  $E = \eta_{eq} N_s$ . Therefore, we substitute  $\eta_{eq} N_s$  into the expression for the Helstrom limit

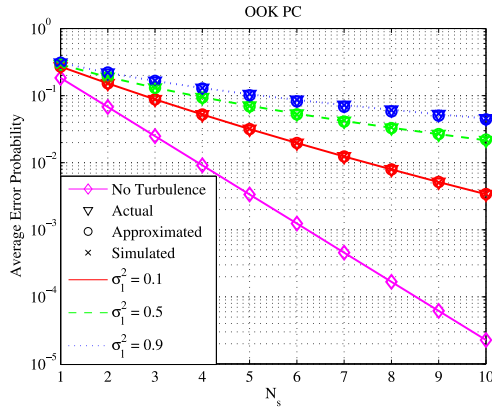
$$P_{\text{BPSK}}^{\text{Hels}}(\eta_{eq}) = \frac{1}{2} \left( 1 - \sqrt{1 - e^{-4\eta_{eq} N_s}} \right). \quad (21)$$

This formula gives the quantum limit for the probability of error in the presence of turbulence. The average error probability is given by

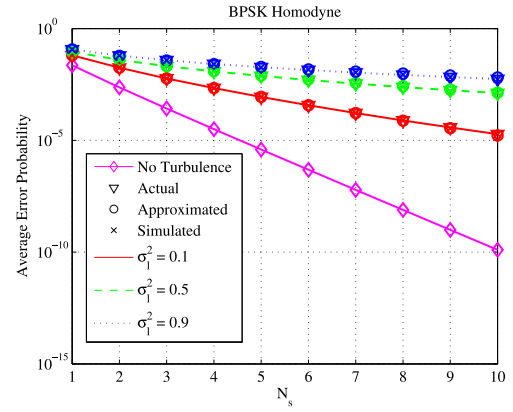
$$\bar{P}_{\text{BPSK}}^{\text{Hels}} = \int_0^\infty \frac{1}{2} \left( 1 - \sqrt{1 - e^{-4\eta_{eq} N_s}} \right) \times \frac{1}{\eta_{eq} \sqrt{2\pi\sigma_l^2}} \exp \left( -\frac{\left( \ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) + \frac{\sigma_l^2}{2} \right)^2}{2\sigma_l^2} \right) d\eta_{eq}. \quad (22)$$

By substituting  $\ln\left(\frac{\eta_{eq}}{\eta_{sc}}\right) = z$  and using the result from [14, eq. (5-7)], the average error probability for BPSK Helstrom detection can be approximated as

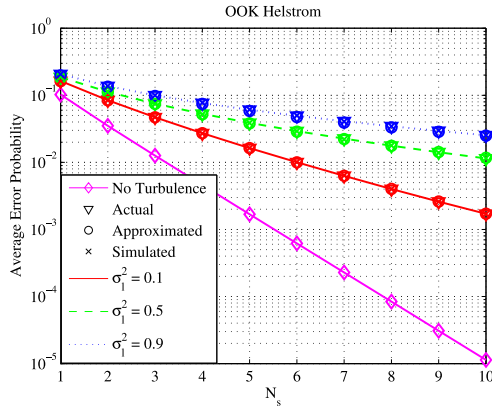
$$\bar{P}_{\text{BPSK}}^{\text{Hels}} \approx \frac{1}{2} - \frac{1}{2} \left[ \frac{2}{3} \sqrt{1 - \exp(-4\eta_{sc} N_s e^{-\sigma_l^2/2})} + \frac{1}{6} \sqrt{1 - \exp(-4\eta_{sc} N_s e^{-\sigma_l^2/2 + \sqrt{3}\sigma_l})} + \frac{1}{6} \sqrt{1 - \exp(-4\eta_{sc} N_s e^{-\sigma_l^2/2 - \sqrt{3}\sigma_l})} \right]. \quad (23)$$



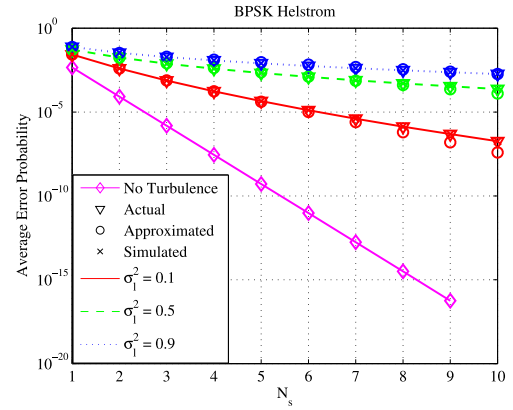
(a) OOK photon counting receiver.



(a) BPSK Homodyne receiver.



(b) OOK Helstrom receiver.



(b) BPSK Helstrom receiver.

Fig. 1. Average error probability of OOK receivers.

Fig. 2. Average error probability of BPSK receivers.

It can be observed that as  $\sigma_l^2$  increases, the quantity inside the exponential function increases, leading to a higher average error probability.

## V. LOWER BOUNDS FOR AVERAGE CAPACITIES

The Shannon capacities for the optimal quantum detection of OOK and BPSK are given as [9]

$$\begin{aligned} C_{\text{OOK}}^{\text{Hels}} &= 1 - h(P_{\text{OOK}}^{\text{Hels}}), \\ C_{\text{BPSK}}^{\text{Hels}} &= 1 - h(P_{\text{BPSK}}^{\text{Hels}}), \end{aligned} \quad (24)$$

where  $h(p) = -p \log_2(p) - (1-p) \log_2(1-p)$  and  $P_{\text{OOK}}^{\text{Hels}}$ ,  $P_{\text{BPSK}}^{\text{Hels}}$  denote the instantaneous error probabilities of OOK Helstrom and BPSK Helstrom detectors, respectively. While initially derived for a binary symmetric channel (BSC), these expressions apply here because Helstrom detection minimizes error probability, making the optical quantum channel behave like a BSC. In the presence of atmospheric turbulence,  $P_{\text{OOK}}^{\text{Hels}}$  and  $P_{\text{BPSK}}^{\text{Hels}}$  become random quantities, and hence these capacities also become random depending on the instantaneous transmissivity of turbulence,  $\eta_t$ . Calculating the exact average capacities is challenging due to the complex relationship between  $C$  and the random channel transmittance  $\eta_t$ . Therefore, we present a lower bound.

Given that the entropy function  $h(p)$  is concave, Jensen's inequality allows us to assert that  $\mathbb{E}[h(p)] \leq h(\mathbb{E}[p])$ , where

$\mathbb{E}[\cdot]$  denotes the expectation operator. So, the lower bounds for the average capacities of OOK with Helstrom detection, and BPSK with Helstrom detection are given by

$$\begin{aligned} \overline{C}_{\text{OOK}}^{\text{Hels}} &\geq 1 - h(\overline{P}_{\text{OOK}}^{\text{Hels}}), \\ \overline{C}_{\text{BPSK}}^{\text{Hels}} &\geq 1 - h(\overline{P}_{\text{BPSK}}^{\text{Hels}}), \end{aligned} \quad (25)$$

where  $\overline{P}_{\text{OOK}}^{\text{Hels}}$  and  $\overline{P}_{\text{BPSK}}^{\text{Hels}}$  are given by (17) and (23) respectively. These bounds provide insights into the ultimate performance limits of optical communication systems under atmospheric turbulence, leveraging quantum detection methodologies. The analytical results derived in this letter can be used for the rational system design of quantum-enhanced classical data transmission in future communication systems.

## VI. RESULTS AND DISCUSSION

The average error probability in the presence of weak atmospheric turbulence, characterized by a lognormal distribution, is computed using our proposed approximate closed-form expression. System performance is analyzed by evaluating the average error probability across various turbulent strengths. We assume a scattering loss of  $\eta_{sc} = 2$  dB. Monte Carlo simulations are conducted with  $10^6$  samples from the lognormal distribution. Fig. 1 shows the average error probability of the OOK receivers as a function of  $N_s$  for different values of  $\sigma_l^2$ . It is observed that the error probability increases

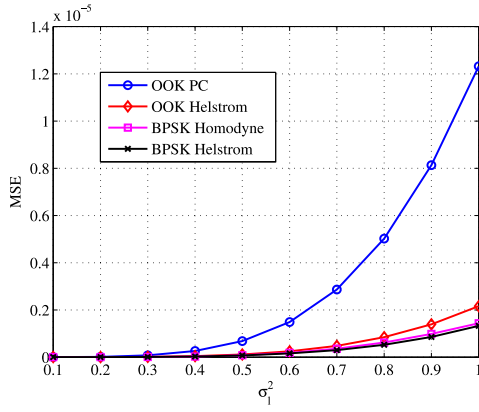


Fig. 3. MSE between actual and approximated error probabilities.

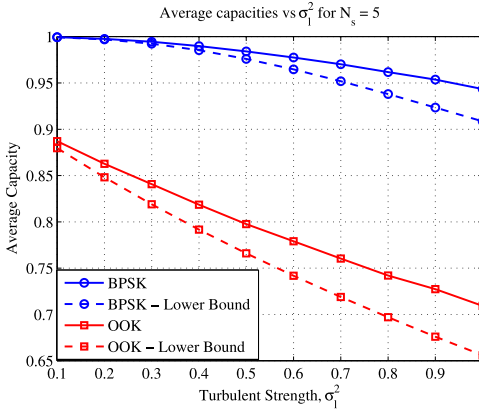


Fig. 4. Average channel capacities of Helstrom detection.

with higher  $\sigma_l^2$ . Notably, the graphs generated using our approximate closed-form expressions closely match the exact expressions (evaluated using numerical integration) and those obtained from numerical Monte Carlo simulations. Fig. 2 presents similar plots for the BPSK receivers.

The mean square error (MSE) is an effective metric for quantifying the accuracy between exact values and approximations used for evaluating the average error probabilities. We compute the MSE between numerically estimated exact values and approximated values across different  $\sigma_l^2$  values and present them in Fig. 3. The results show that the values computed using our approximation closely align with the exact values, with the worst-case MSE being less than  $1.3 \times 10^{-5}$ .

Fig. 4 depicts the average capacities of Helstrom detectors as a function of turbulent strength for  $N_s = 5$ . The average capacities of (24) are obtained using Monte Carlo simulations, while the lower bounds are plotted using the closed-form expressions from (25). It is evident that the average capacity monotonically decreases with increasing turbulent strength, indicating a reduction in the rate at which the sender can reliably transmit classical information through the channel. Furthermore, the impact of  $\sigma_l^2$  on the average capacity of OOK is greater compared to that of the BPSK capacity.

## VII. CONCLUSION

In this letter, we have derived approximate closed-form expressions for average error probabilities in weakly turbulent free-space optical quantum communication channels for both classical and quantum detection scenarios. Our analysis focuses on binary discrete modulations OOK and BPSK transmitted over a log-normal turbulent channel under conditions of weak-to-moderate atmospheric turbulence. The accuracy of our proposed expressions is demonstrated through their close agreement with exact values obtained from numerical methods. From these error probabilities, we have established lower bounds for average capacities of Helstrom detection, providing insights into the fundamental limits of communication under turbulent environmental conditions. These findings underscore the practical relevance and applicability of our analytical framework, paving the way for enhanced design and optimization of quantum communication systems in adverse atmospheric conditions.

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